DS_Day01_22

22. AVIATION — WHY ARE FLIGHTS DELAYED?

Industry: Aviation

An airline wants to reduce flight delays. Students receive historical flight-level data containing airport, aircraft, scheduled departure, actual departure, weather and previous flight information. Students must calculate delay rates by airport. They should analyse delays by hour and day. They should investigate the effect of aircraft turnaround time. They should determine whether certain routes consistently experience delays. They should distinguish operational delays from weather-related delays. They should identify the most significant delay contributors. The final output should focus on operational recommendations.

Data provided

Flight_ID, Airport, Route, Aircraft, Scheduled_Departure, Actual_Departure, Weather, Previous_Flight, Turnaround_Time

Student Tasks

• Clean and validate the provided dataset

Perform exploratory data analysis using suitable descriptive statistics

Create clear visualisations for the important patterns in the data

Students receive historical flight-level data containing airport, aircraft, scheduled departure, actual departure, weather and previous flight information

Calculate delay rates by airport

Analyse delays by hour and day

Investigate the effect of aircraft turnaround time

Determine whether certain routes consistently experience delays

Distinguish operational delays from weather-related delays

Identify the most significant delay contributors

Specified Machine Learning Algorithm

Random Forest

Expected Output

5 to 7 key insights, 4 visualisations, one trained Random Forest model, model evaluation results and a practical action plan.